BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, BANNU

INTERMEDIATE (11TH CLASS) EXAMINATION, 2021

MATHEMATICS

Past Paper 2021

Retyped faithfully from the original board paper — same questions, no changes.

Paper Maths - 11th — Annual Exam – 2021 — SECTION-A

Time Allowed: 20 mints | **Maximum Marks:** 20

Note: Note: Q.1: Fill in the blanks with sutable option. Student ticks/underlines ignored. Q4(C)(D) numerators read under watermark as x (best read).

Q1. Period of sec (3x) is.....

(A) π (B) 0

(C) 2π (D) 2π/3

Q2. The graph of $y = 2x^2 + 1$ is.....

(A) Parabola (B) circle

(C) straight line (D) none

Q3. The amplitude of 5sin (2x) is.....

(A) 1 (B) -5

(C) 5 (D) 10

Q4. The inverse function of f(x) = x/(x-1) is.....

(A) (x-1)/x (B) (x+1)/(x-1)

(C) x/(1-x) (D) x/(x-1)

Q5. When x = 5 then 7!(x+6) / x!(x+1)! = ...

(A) 36 (B) 42

(C) 7 (D) 7!

Q6. 1 - x + x2 - x3 + ... is the expension of.....

(A) $(1 - x)^{-2}$ (B) $(1 - x)^{-1}$

(C) $(1 + x)^{-2}$ (D) $(1 + 2)^{-1}$

Q7. If $\vec{c}$ lies in the plane of the $\vec{a}$ and $\vec{b}$, then $\vec{c}$ = ...

(A) $\alpha\vec{a} + \beta\vec{b}$ (B) $\alpha\vec{a} \times \beta\vec{b}$

(C) $\alpha\vec{a} \cdot \beta\vec{b}$ (D) $\vec{0}$

Q8. The domain of $f(x) = \sqrt{(x + 1)}$ is the set of all.....

(A) $X > 0$ (B) $x > 1$

(C) $x \geq 1$ (D) None

Q9. The solution of linear inequality $3x - 2 \geq 8 + 5x$ is.....

(A) $[-\infty, -5)$ (B) $(-\infty, -5]$

(C) $[-\infty, 5]$ (D) None

Q10. In right angled $\triangle ABC$ $\alpha = 57°$ then β = ...

(A) 57° (B) 90°

(C) 33° (D) 22°

Q11. The multiplicative inverse of $Z = 2 + 3i$ is.....

(A) $2 - 3i$ (B) $-2 - 3i$

(C) $2/13 - 3/13\ i$ (D) $2/13 + 3/13\ i$

Q12. Let A = [[3,-2],[5,7]] then the co-factor of -2 is .....

(A) -5 (B) 2

(C) -2 (D) 5

Q13. The direction cosines of $\vec{a} = 2i - j + k$ are .....

(A) 2,1,1 (B) 2,-1,1

(C) $2/\sqrt{6}, -1/\sqrt{6}, 1/\sqrt{6}$ (D) None

Q14. Let $A^2 = A$ then $\Sigma(j=1 \text{ to } 3)\ A^j$ = ...

(A) A (B) 2A

(C) A^3 (D) 3A

Q15. $\Sigma(k=1 \text{ to } n)\ K^2$ = ...

(A) $n(n+1)/2$ (B) $n(n+1)(2n+1)/6$

(C) $[n(n+1)/2]^2$ (D) None

Q16. A fair die is rolled, the probability that even integer will occur is .....

(A) 1/2 (B) 1/3

(C) 1/4 (D) 1/6

Q17. The A.M between 6th and 8th term of the A.P 1, 4, 7, 10 .....is.....

(A) 16 (B) 22

(C) 19 (D) 48

Q18. The vector perpendicular to -I + 2j + k and 3i - k is...

(A) 5i - j - 3k (B) -5i - j - 3k

(C) -5i + j + 3k (D) None

Q19. If p/q is the third term of H.P, then theird term of corresponding A.P will be.....

(A) P (B) q/p

(C) U (D) -p/q

Q20. Domin of sin 2x is .....

(A) R (B) R - {2}

(C) 2π (D) None

SECTION-B

Time Allowed: 2.40 Hrs. | **Maximum Marks:** 50

Q2. (Marks 50) Q.2: Attempt any Ten (10) parts. All parts carry equal marks.

i. If $Z_1 = 1 + 3i$, $Z_2 = 2 + 3i$ evaluate $|Z|_1/|Z|_2$

ii. For what value of λ the homogeneous system x + 5y + 3z = 0, 5x + y - λz = 0, x + 2y + λz = 0 has non-trivial solution.

iii. Solve the triangle for which a = 9, b = 7, c = 5.

iv. Find the unit vector perpendicular to both $\vec{a}$ = i + j + 2k and $\vec{b}$ = -2i + j - 3k

v. For a G.P $a_{10} = l$, $a_{13} = m$, $a_{16} = n$ show that $l\,n = m^2$

vi. Give $a_n = 201$ for the A.P 5, 9, 13,..... find n.

vii. Solve for 'n' when ${}^{n+1}C_4 = 6\,{}^{n-1}C_2$.

viii. Simplify $\Sigma(k=0 \text{ to } 3)\ 2^k/(k+1)$

ix. Prove that $2n > n\ \forall\ n \in N$

x. Find the area of the inscribed circle of the triangle with measures of the sides 55m, 25m, 70m.

xi. Prove that identity $[\sin(\alpha)+\sin(9\alpha)]/[\cos(\alpha)+\cos(9\alpha)] = \tan(5\alpha)$

xii. If A = |1 2; -1 3|, show that $(A^{-1})^{-1} = A$.

xiii. Graph y = -cos (x) in $0 \le x \le 2\pi$

SECTION-C

Maximum Marks: 30

> Note: Note: Attempt any Three (3) of the following questions, All questions carry equal marks. Q.4(a) printed as (1+4)^(1/4) — verbatim as on scan (likely board misprint of (1+x)^(1/4)).

Q3.

a) Find the period, frequency, maximum and minimum value of the function $y = 5 - 4 \sin(\theta + 30)$

b) Show that $\tan^{-1} 3/4 + \tan^{-1} 8/19 = \pi/4$

Q4.

a) If x4 and higher powers be neglected, then show that (1+4)^(1/4) + (1-x)^(1/4) = $a - bx^2$. Find also a and b.

b) Show by mathematical induction that $(3^{2n} \cdot 2^{2n})/5$ is an integer for all positive integers.

Q5.

a) Prove that ${}^{n}C_r + {}^{n}C_{r+1} = {}^{n+1}C_r$

b) What is the smallest angle of a triangle whose sides measure 25, 18 and 21ft.

Q6.

a) Solve the system of equation 2x + 4y - z = 10, x - 2y - 2z = 4, 3x + y + z = 7 by Gauss elimination method.

b) Show that $f(x) = x\sqrt{(x^2 + 3)}$ is odd function.